7.2 Stretching Without Turning

WORKSHEET

The question

The transformation T from $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} X \\ Y \end{pmatrix}$ is given by

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

Show that T leaves the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ unchanged in direction but multiplied by a scalar, and that $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is similarly transformed.

The circle C whose equation is $x^2 + y^2 = 1$ transforms under T to a curve E. Show that E has equation

$$8X^2 + 12XY + 17Y^2 = 80,$$

and state the area of the region bounded by E. Show also that the greatest value of X on E is $2\sqrt{17/5}$.

Find the equation of the tangent to E at the point which corresponds to the point $\frac{1}{5}(3,4)$ on C.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What does it mean for T to leave a vector "unchanged in direction but multiplied by a scalar"? Write it as an equation.

STAGE 2 The image of the circle

Ask yourself To find the equation of E, express x and y in terms of X and Y (invert the matrix), and substitute into $x^2 + y^2 = 1$.

Ask yourself What shape is E, and what is its area? Use the eigenvectors.

STAGE 3 The greatest value of X

Ask yourself On C, $X = \frac{2}{5}(9x - 2y)$. How large can $9x - 2y$ be when $x^2 + y^2 = 1$?

STAGE 4 A tangent

Pause and try The point $\frac{1}{5}(3,4)$ on C has tangent line $3x + 4y = 5$. A linear map sends tangent lines to tangent lines. Use this to find the tangent to E.

Try this next

1. **Finding eigenvalues.** Show that the eigenvalues of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are the roots of $\lambda^2 - (a + d)\lambda + (ad - bc) = 0$, and check that the matrix in the question gives $\lambda = 2$ and 4.

2. **Perpendicular eigenvectors.** Show that if $\mathbf{M}$ is symmetric and $\mathbf{Mu} = \lambda\mathbf{u}$, $\mathbf{Mv} = \mu\mathbf{v}$ with $\lambda \neq \mu$, then $\mathbf{u} \cdot \mathbf{v} = 0$. (*Hint:* compute $\mathbf{v} \cdot \mathbf{Mu}$ in two ways.)

3. **A rotation.** Show that the rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ has no real eigenvectors unless θ is a multiple of π. What are its complex eigenvalues?

4. **Powers.** Using the eigenvectors, find $T^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. (*Hint:* write $(1,0)$ as a combination of $(1,2)$ and $(2,-1)$.)

Further reading

- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Chapter 6 is an excellent introduction to eigenvalues and their applications.
- **Kurt Bryan & Tanya Leise, "The $25,000,000,000 eigenvector: the linear algebra behind Google"**, *SIAM Review* 48 (2006). Readable and freely available.
- **3Blue1Brown, *Essence of Linear Algebra*** (video series, free online). Beautiful animations of exactly the geometric ideas in this section.

The full guided solution, and what the question leads to at university, is in Section 7.2:
https://beyondstep.keithmcnulty.org/ch07/eigenvectors.html